Curriculum Vitae

Xinyi Zhao

Email: zhao@demogr.mpg.de / zhao@mpib-berlin.mpg.de
Phone: (+49) 151 6715 0711
ORCID: 0000-0002-2552-7795
Address: Adaptive Rationality Center, Max Planck Institute for Human development, Lentzeallee 94, 14195 Berlin, Germany

Research Interests

Computational Social Science	Digital and computational demography
Gender disparities	Migration studies
Science of Science	Applied statistics
Cartography	Geographic Information System (GIS)

Education

2021.10 - 2024.08 **Doctorate of Philosophy in Sociology (pass with no corrections)**
Department of Sociology and St Hugh's College, University of Oxford, Oxford, United Kingdom
- Thesis title: Gender inequality in academia: Opportunities and challenges in the era of globalization and digitalization
- Jointly affiliated with International Max Planck Research School on Population Health and Data Science (IMPRS-PHDS)

Supervisor: Prof. Ridhi Kashyap (Oxford)

2020.09 - 2024.08 **Doctoral researcher**
Laboratory of Digital and Computational Demography and IMPRS-PHDS
Max Planck Institute for Demographic Research, Rostock, Germany
Supervisor: Prof. Emilio Zagheni

2016.09—2019.06 **Master of Science in Cartography and Geographical Information System**
Nanjing University, Nanjing, China (GPA: 90.28/100)
Thesis title: Spatio-temporal Dynamics Analysis of Interprovincial Migration System in China
Supervisor: Prof. Yingxia Pu

2012.09—2016.06 **Bachelor of Science in Geographical Information System**
China University of Geosciences, Wuhan, China (GPA: 3.64/5, Rank: 3/96)
Thesis title: Modelling OD Flows of Bus Trip based on GPS and IC Card Data

Work Experience

2024. 09 - **Adaptive Rationality Center, Max Planck Institute for Human Development**
Postdoc Researcher

2023.12 -2024.09 **Department of Sociology and Leverhulme Centre for Demographic Science, University of Oxford**

Research Assistant
Main responsibilities: spatial data collection and processing

2020.01 - 2020.09 **Department of Urban planning and Design, The University of Hong Kong**
Full-time Research Assistant
Main responsibilities: data collection and analysis; model programing; project reporting

Research Stays

2026.05 **Northwestern Institute on Complex Systems (NICO)**
Visiting researcher

2026.02 **Swedish Excellence Center for Computational Social Science, Linköping university**
Junior visiting fellow

2025.10 **Institute of Human-Centred Computing, Graz University of Technology**
Visiting researcher

2024.05 **CEWS Center of Excellence Women and Science, GESIS – Leibniz Institute for the Social Sciences, Köln**
Visiting researcher

2023.09 **Mannheimer Zentrum für Europäische Sozialforschung (MZES)**, **University of Mannheim**
Visiting fellow

Publications

Journal:

1. **Zhao, X.**, Derrick, G., Larivière, V., and Sugimoto, C. R. (2026). "Gender Differences in the Conflict between the Beginning of an Academic Career and Parenthood". (***Accepted by Higher Education***) ***SocArXiv.*** https://doi.org/10.31235/osf.io/ksd6q_v1
   - Awarded by Best Poster Award at the Annual Academy of International Max Planck Research School for Population, Health and Data Science

2. Breen, C.F, Fatehkia, M., Yan, J., **Zhao, X.**, Leasure, D.R., Weber, I., and Kashyap, R. (2025). "Mapping subnational gender gaps in internet and mobile adoption using social media data", ***Proceedings of the National Academy of Sciences (PNAS).*** https://doi.org/10.1073/pnas.2416624122

3. Jacobs, E., Theile, T., Perrotta, D., **Zhao, X.**, Anastasiadou, A., and Zagheni, E. (2025). "Global Gender Gaps in the International Migration of Professionals on LinkedIn", ***Population and Development Review.*** http://doi.org/10.1111/padr.70012

4. Akbaritabar, A., Dańko, M.J., **Zhao. X.,** and Zagheni, E. (2025). "Global subnational estimates of migration of scientists reveal large disparities in internal and international flows", ***Proceedings of the National Academy of Sciences (PNAS)***. https://doi.org/10.1073/pnas.2424521122

5. Yin, Y., Pu, Y., **Zhao, X.**, Chi, G., and Ye, C. (2023). "The Spatiotemporal Elasticity of Age Structure in China's Interprovincial Migration System" ***Sustainability***. https://www.mdpi.com/2071-1050/15/10/8001

6. **Zhao, X**., Akbaritabar, A., Kashyap, R., and Zagheni, E. (2023). "A gender perspective on the global migration of scholars" ***Proceedings of the National Academy of Sciences (PNAS)***. https://doi.org/10.1073/pnas.2214664120

- Awarded Jan Hoem award for best paper by Max Planck Institute for Demographic Research
- Media coverage in Nature, Times Higher Education Supplement, PNAS Science Sessions Podcasts, e.g.,

PNAS Science Sessions Podcast on "Gender gap among migrant scientists". ***PNAS*** (July 2023)
More female researchers than ever migrate internationally. ***Nature*** (March 2023)
Wereldwijde mobiliteit onder vrouwelijke wetenschappers versnelt. ***ScienceGuide*** (March
Female academics migrate shorter distances and to fewer countries than men, finds study. ***Physicsworld*** (March 2023)
Female scienists miss out on overseas opportunities. ***Nature.com*** (March 2023)
Female researchers close global mobility gap on men. ***Times Higher Education*** (February 2023)
A gender perspective on the global migration of scholars. ***sciencenewsnet.in*** (February 2023)
A gender perspective on the global migration of scholars. ***phys.org*** (February 2023)
A gender perspective on the global migration of scholars. ***newswise*** (February 2023)

7. **Zhao, X.**, Aref, S., Zagheni, E., and Stecklov, G. (2022). "Return migration of German-affiliated researchers: Analyzing departure and return by gender, cohort, and discipline using Scopus bibliometric data 1996-2020" ***Scientometrics.*** https://doi.org/10.1007/s11192-022-04351-4

- Media coverage in Population Europe: Will we ever see them again? ***Population Europe*** (Jul 2022)

8. **Zhao, X**., Aref, S., Zagheni, E., and Stecklov, G. (2021). "Mapping academic migration of German-affiliated researchers across countries using 8 million Scopus publications from 1996 to 2020", ***Abstr. Int. Cartogr. Assoc.***, 3, 327. https://doi.org/10.5194/ica-abs-3-327-2021

9. Pu, Y., **Zhao, X.**, Chi, G., Zhao, J., and Kong, F. (2019). "A Spatial Dynamic Panel Approach to Modelling the Space-time Dynamic of Interprovincial Migration Flows in China" ***Demographic Research*** 41(31): 913-948. https://doi.org/10.4054/DemRes.2019.41.31

10. Pu, Y., **Zhao, X.**, Chi, G., Zhao, S., Wang, J., Jin, Z., and Yin, J. (2019). "Design and Implementation of a Parallel Geographically Weighted k- Nearest Neighbor Classifier" ***Computers and Geosciences*** 6(127): 111-122. https://doi.org/10.1016/j.cageo.2019.02.009

11. **Zhao, X**., and Pu, Y. (2018). "Space-time Spillover Effects and Driving Forces of Regional Migration Process" ***Journal of Geoinformation Science*** 6(29): 817-826 (in Chinese).

12. PU, Y., Deng, D., and **Zhao, X.** (2018). "The Measurement of New Urbanization Level in Sichuan Province Based on AHP-BN Methods" ***Journal of Northwest Normal University*** (Natural Science) 1(54): 77-83 (in Chinese).

Proceeding:

1. **Zhao, X.**, Aref, S., Zagheni, E., and Stecklov, G. (2021). "International migration in academia and citation performance: an analysis of German-affiliated researchers by gender and discipline using Scopus publications 1996-2020" In: ***Proceedings of the 18th International Conference on Scientometrics and Informetrics (ISSI 2021)***, Leuven, Belgium, July 12-15, 2021. Leuven: ISSI. https://arxiv.org/abs/2104.12380

2. **Zhao, X**., and Pu, Y. (2018). "The Complex System of Interprovincial Migration Flows in China, 1985–2015" 2018 26th ***International Conference on Geoinformatics***, 2018: 1-6. https://doi.org/10.1109/GEOINFORMATICS.2018.8557095

Working papers

1. Lin, Y., Gu, H., **Zhao, X.**, Xu, Z., Wang, Y., Sugimoto, C., and Min, C. "Sino-US Trade War has hindered bilateral scientific mobility" (***in progress***)

2. **Zhao, X.**, Akbaritabar, A., Kashyap, R., and Zagheni, E. "Gender differences in online visibility of early career researchers". (***Resubmit to Nature Communications***)

   - Awarded by 3rd place Best Poster Award and audience favorite award at Anniversary Student Poster Competition of Sociology department, University of Oxford

3. **Zhao, X.,** Hertwig, R., and Wulff, D**. "**Sustainable Development Goals in Psychology: A Century of Progress in Publications**"** (***R&R, American Psychologist***) ***ArXiv.*** https://doi.org/10.48550/arXiv.2512.08628

4. **Zhao, X**., Thoma, A.I, Hertwig, R., and Wulff, D.U. (2025). "Mapping the gender attrition gap in academic psychology", (***Under review, American Psychologist***) ***ArXiv.*** https://doi.org/10.48550/arXiv.2510.13273

5. **Zhao, X.,** and Wulff, D**. "**An LLM-Assisted Survey for Gendered Pathways and Attrition in Academic Careers**"** (***in progress***)

6**. Zhao, X.,** and Benson Malinga John**. "**Polygyny–Fertility Nexus Revisited in sub-Saharan Africa: Is it Real or Spurious?**"** (***in progress***)

7. **Zhao, X.,** and Jiaxin Shi**. "**Internal migration of researchers**"** (***in progress***)

8. **Zhao, X.,** and Shuye Yu**. "**The Causal Link Between Maternal Travel Burden and Newborn Health**"** (***in progress***)

Book Chapters

1. Kashyap, R.; Rinderknecht, R. G.; Akbaritabar, A.; Alburez-Gutierrez, D.; Gil-Clavel, B. S.; Grow, A.; Kim, J.; Leasure, D. R.; Lohmann, S.; Negraia, D. V.; Perrotta, D.; Rampazzo, F; Tsai, C.-J.; Verhagen, M. D.; Zagheni, E.; **Zhao, X**. (2023). Digital and computational demography
In: ***Research handbook on digital sociology***, 48–86. Cheltenham: Edward Elgar Publishing.

International Conferences/workshops

2025:
Workshop: Women and Work in the 21st Century. Berlin, Germany. September 11-12.
Oral presentation: Digital Inequality in Academia.

11th International Conference on Computational Social Science (IC2S2). Norrköping, Sweden. July 21-24. Poster presentation: 1) Mapping the Leaky Pipeline in Psychology: Analysis of Gender Disparities in Academic Progression; 2) Sustainable Development Goals in Psychology: A Century of Progress in Publications.

International Conference on the Science of Science and Innovation (ICSSI). Copenhagen, Denmark. June 16-18. Poster presentation: Mapping the Leaky Pipeline in Psychology: Analysis of Gender Disparities in Academic Progression.

Bridging Approaches in the Sciences Studying Science. Munich, Germany. February 18-23. Oral presentation: Mapping the leaky pipeline: Analysis of gender disparities in academic progression within psychology.

2024:
Digital Inequalities Symposium for Early-Career Researchers. Paris, France. December 19. Oral presentation: Gender Difference differences in the online visibility of early-career researchers.

Scholarly Migration and Mobility Symposium. Rostock, Germany. October 15. Poster presentation: Gendered patterns of global scholarly migration

Workshop on Digital Technologies and Sustainable Development. Oxford, UK. September 30-October 01.

*31st Nordic Association Conference***.** Norrköping, Sweden. August 14-16. Oreal presentation: Gender Difference differences in the online visibility of early-career researchers.

European Population Conference (EPC) 2024. Edinburgh Scotland, UK. June 12 – 15. Flash talk: Gender in the Conflict between the Beginning of an Academic Career and Parenthood.

Gender Summit 2024. Online. May 22 - 23. Oral presentation: A gender perspective on the global migration of scholars.

RC28 Spring Meeting 2024. Shanghai, China. May 17 - 19. Oral presentation: Gender in the Conflict between the Beginning of an Academic Career and Parenthood.

2024 Population Association of America (PAA). Columbus Ohio, US. April 17 - 20.
Flash talk:
Gender Difference differences in the online visibility of early-career researchers.
Gender Difference in Trajectories of Childbearing and Doctoral Degree Pursuit Among Researchers.

2023:
Quetelet 2023 Seminar: Unconventional Data Sources for Population. Louvain-la-Nauve, Belgium. November 09 – 10. Oral presentation: Gender differences in the online visibility of early-career researchers.

The Third Asian Population Forum. Shanghai, China. October 31 – November 01. Oral presentation: Gendered patterns of global scholarly migration: Evidence on mobile Researchers from bibliometric data.
knowledge Societies – 4th Conference of the Academy of Sociology. Bern, Switzerland. August 28 - 30. Oral presentation: Gender differences in the online visibility of early-career researchers: Are men more likely to gain mentions on Twitter and benefit more?

The 9th International Conference on Computational Social Science (IC2S2). Copenhagen, Denmark. July 17 – 20. Poster presentation: Gender differences in the online visibility of early-career researchers.

knowledge Societies – 4th Conference of the Academy of Sociology. Bern, Switzerland. August 28 - 30. Oral presentation: Gender differences in the online visibility of early-career researchers: Are men more likely to gain mentions on Twitter and benefit more?

The 20th IMISCOE Annual Conference. Warsaw, Poland. July 3 – 6. Oral presentation: Gendered patterns of global scholarly migration: Evidence on mobile researchers from bibliometric data.

The 2nd MigrantLife Symposium. St-Andrews, UK. June 29-30. Oral presentation: Gendered patterns of global scholarly migration: Evidence on mobile researchers from bibliometric data & Case study in moving-out and return researchers with German affiliations.

knowledge Societies – 4th Conference of the Academy of Sociology. Bern, Switzerland. August 28 - 30. Oral presentation: Gender differences in the online visibility of early-career researchers: Are men more likely to gain mentions on Twitter and benefit more?

2023 Population Association of America (PAA). New Orleans, USA. April 12 - 15. Poster presentation: Gendered patterns of global scholarly migration: Evidence on mobile researchers from bibliometric data.

The workshop on "Detecting and Mitigating Bias in Migration Research". Córdoba, Spain. March 30 - 31. Oral presentation: Identifying and Mitigating Bias in the Study of Scholarly Migration using Bibliometric Data

Popdays 2023- Giornate di Studio sulla Popolazione. Rome, Italy. Feb 01 - 03. Oral presentation: Gender differences in the online visibility of early-career researchers: Are men more likely to gain mentions on Twitter and benefit more?

2023 Population Association of America (PAA). New Orleans, USA. April 12 - 15. Oral presentation: Gendered patterns of global scholarly migration: Evidence on mobile researchers from bibliometric data.

2022:

The 28th Annual Population Postgraduate Conference. Florence, Italy. October 26 - 28. Oral presentation: Gendered patterns of global scholarly migration: Evidence on mobile Researchers from bibliometric data.

VI Portuguese Conference of Demography. Lisbon, Portugal. October 13 - 14. Oral presentation:
Gendered patterns of global scholarly migration: Evidence on mobile researchers from bibliometric data.

British Society for Population Studies 2022 Conference. Winchester, UK. September 05 - 07. Oral presentation: Gendered patterns of global scholarly migration: Evidence on mobile researchers from bibliometric data.

2022 Annual Meeting of the American Sociological Association. Los Angeles, USA. October 13 - 14. Oral presentation:
Gendered patterns of global scholarly migration: Evidence on mobile researchers from bibliometric data.

European Population Conference (EPC) 2022. Groningen, Netherland. June 29 - July02. Oral presentation: Return migration in academia: an analysis of German-affiliated researchers by gender, cohort and discipline using Scopus publications 1996-2020.

2022 Population Association of America (PAA 2022). Atlanta, USA (online). April 06-09. Oral presentation: Gender Inequality in International Scholarly Migration: Evidence on German-Affiliated Researchers From 8 million Scopus Publications.

2021:

The 30th International Cartographic Conference T. Florence, Italy (online). December 14-18. Oral presentation: Gendered patterns of global scholarly migration: Mapping academic migration of German-affiliated researchers across countries using 8 million Scopus publications from 1996 to 2020.

2021 International Population Conference (IPC). Hyderabad, India (online). December 5-10. Oral presentation: The interplay between scholarly migration, gender, career stages, and citations: Evidence on German-affiliated researchers from 8 million Scopus publications.

Migration and Mobility Research in the Digital Era (MIMODE 2021). Lyon, France (online). October 27. Oral presentation: The interplay between migration, gender, career stages, and citations: Evidence on German-affiliated researchers from Scopus publications.

Workshop "Methodological Innovations in Research on Spatial Mobility: Potentials and Pitfalls. Berlin, Germany (online). September 27. Oral presentation: The interplay between migration, gender, career stages, and citations: Evidence on German-affiliated researchers from Scopus publications.

Flash Talk Competition at the Conference "New Data for the New Challenges of Population and Society ": 3-min talk. Padua, Italy (online). September 22-24. Oral presentation: The interplay between scholarly migration, gender, career stages, and citations: Evidence on German-affiliated researchers from 8 million Scopus publications.

The 7th International Conference on Computational Social Science (IC2S2). Zurich, Switzerland (online). July 27-31. Oral presentation: The interplay between migration, gender, career stages, and citations: Evidence on German-affiliated researchers from Scopus publications.

The 18th International Conference on Scientometrics & Informetrics (ISSI). Leuven, Belgium (online). July 12-15. Oral presentation: International migration in academia and citation performance: an analysis of German-affiliated researchers by gender and discipline using Scopus publications 1996-2020.

2018:

The 58th European Regional Science Association (ERSA) Congress. Cork, Ireland. August 28-31. Oral presentation (Young Scientist Session): Modeling and Interpreting the Space-time Dynamics of Migration System: A case of Interprovincial Migration in China, 1985-2015.

The 4th Asian Population Association (APA) Conference. Shanghai, China. July 11-14. Oral presentation: Modeling and Interpreting the Space-time Dynamics of Migration System: A case of Interprovincial Migration in China, 1985-2015.

The 26th International Conference on Geoinformatics. Kunming, China. June 28-30. Oral presentation: Modeling and Interpreting the Space-time Dynamics of Migration System: A case of Interprovincial Migration in China, 1985-2015.

2017:

The Asia GIS Conference 2017. Hong Kong, China. January 5-7. Oral presentation: Design and implementation of parallel geographically weighted k-nearest 2017 neighbor classifier.

2016:

The 33rd International Geographical Congress. Beijing, China. August 21-25. Poster presentation: The design and implementation of parallel gwk-NN classifier.

Awards and Honors

2024:

3rd Place and audiences' favorite poster at Anniversary Student Poster Competition of Department of Sociology, University of Oxford

2023:

Jan M. Hoem Paper Award, awarded to best paper by early career researcher in Max Planck Institute for Demographic Research

Best Poster Award at the Annual academy of International Max Planck Research School for Population, Health and Data Science

2021:

Student Travel Award for the 18th International Conference on Scientometrics & Informetrics (ISSI)

International Max Planck Research School for Population, Health and Data Science / Departmental/ Leverhulme Centre for Demographic Science Scholarship

2013-2019:

First prize of Excellent Master Thesis of Surveying and Mapping in Jiangsu province (2019)

First prize of Excellent Paper at China Annual Conference on Theories and Methods of Geographic Information Science（2017）

Second Prize Scholarship at Nanjing University (2018, 2017)

Second prize of Excellent Postgraduates' Thesis at Geography Postgraduates Forum in Jiangsu province (2016)

First Prize Scholarship at Nanjing University (2016)

Outstanding Graduate at China University of Geosciences (2016)

Grand prize of the 14th Challenge Cup National College Student Curricular Academic Science and Technology Works Completion in Hubei province (2015)

Presidential Scholarship at China University of Geosciences (2014, 2013)

Outstanding works award of the Winter Vacation Social Practice at China University of Geosciences (2014)

Second prize of College Physics Competition at China University of Geosciences (2014)

Third prize of Golden Idea Competition at China University of Geosciences (2013)

Academic Memberships

1. Nature Humanities & Social Sciences Editorial Community
2. Demographic Science Unit and Leverhulme Centre for Demographic Science (LCSD) Affiliate Membership
3. Max Planck Institute for demographic Research Affiliated Researcher
4. IMSCOE (International Migration Research Network) PhD Academy
5. International Union for the Scientific Study of Population (IUSSP)
6. Population Association of America (PAA)
7. Migration Oxford

Teaching Service

1. Teaching the session "Migration of Scientists" on course " Migration & Society"
   University of Rostock & MPIDR, 2024

2. Co-organizing a preconference training workshop on: "Using bibliometric data in demographic research"
   European Population Conference, 2022

3. Teaching a training workshop on "Spatial Data Analysis"
   10th Forum on Spatially Integrated Humanities and Social Science, 2019

Reviewing Service

Peer reviewing for ***Population and Development Review*** (2025), ***Quantitative Science Studies*** (2024,2025), ***Comparative Migration Studies*** (2024), ***F1000Research*** (2024), ***Information Processing and Management*** (2023), **Higher education policy** (2022), **The Web Conference 2021** (sub-reviewer)

Additional Training

2022:

Advanced Methods for Causal Inference	Max Planck Institute for Demographic Research
12th Winter School on Longitudinal Social Network Analysis	University of Linköping

2021:

European summer school for Scientometrics 2021	University of Vienna
Data visualization – the art/skill cocktail	Max Planck Institute for Demographic Research
Online webinar: Applied Regression Modeling	University of Greifswald
Population Grids & their Integration with Remote Sensing Data National Aeronautics and Space for Sustainable Development & Disaster Management	Earth Science Applied Science (NASA)

2020:

Demographic Methods	Max Planck Institute for Demographic Research

Digital and Computational Demography	Max Planck Institute for Demographic Research
Population Health	Max Planck Institute for Demographic Research
Basic Statistics for Demographers	European Doctoral School of Demography
Professional Development for Presentation	Max Planck Institute for Demographic Research

2017:

Understanding Complexity	University of Michigan-Peking University Joint Institute
Advanced Quantitative Methodology	University of Michigan-Peking University Joint Institute

Certificates

1. Information system project manager (senior)
2. Software developer
3. MapGIS development engineer

Computer and Language Skills

Proficient in programming (Python, R, C++, C#, JavaScript, Matlab)
Experienced in using professional software (ArcGIS, MapGIS, Erdas, GeoDa, SPSS, Flex)
Languages: Proficiency in written and verbal English; native in Chinese

Referees

Prof. Dr. Emilio Zagheni
Professor and Institute Director, Max Planck Institute for Demographic Research
zagheni@demogr.mpg.de

Prof. Dr. Ridhi Kashyap
Professor of Demography & Computational Social Science, University of Oxford
ridhi.kashyap@nuffield.ox.ac.uk

Dr. Samin Aref
Assistant professor Department of Mechanical & Industrial Engineering, University of Toronto
aref@mie.utoronto.ca

Dr. Aliakbar Akbaritabar
Assistant Professor of Computational Social Science at the Institute of Sociology and Demography of the University of Rostock
akbaritabar@demogr.mpg.de